BIT503 Fundamentals of Information Technology
Assessment 1

Weighting

30%

Learning outcomes

1 Explain the processes associated with information management, including the secure storage and retrieval of data, and apply to the IT workplace using ethical principles.
2 Apply security concepts, tools, and techniques to securely manage hardware, software and networking resources required by an organisation and their end users.

Instructions

Complete and submit your assessment according to the Open Polytechnic's Assessments webpage. This includes information on academic integrity, word limits and referencing.

- Include your name, student number and the assessment number.
- Number your pages.

Submission

- Submit your assessment in one Microsoft Word document.
- Submit your work through your iQualify course.
- Emailed assessments will not be accepted.
- You will receive an automated notice following submission.

By submitting your assessment, you confirm that it is your own, original work.

© The Open Polytechnic of New Zealand

Instruction

Read the case study and answer the questions. Your answers, where necessary, should be evidence-based and appropriately cited and referenced.

Case study

It has become very common for fitness centres to have membership programmes. For example, FitLife has the FitRewards programme, and HealthZone has the ZonePoints programme. These membership programmes are designed to improve customer satisfaction and increase customer commitment to a brand.

While different membership programmes function slightly differently, there are common features:

- Members are offered discounts on certain classes and services. These discounts tend to vary from week to week.
- Members gain points towards bonuses, such as free classes, personal training sessions or merchandise.
- If members register their accounts, weekly newsletters are sent via email to advertise the weekly specials and class schedules.
- Members who have registered their accounts receive additional emails advertising classes and services based on their activity history.

Aside from the benefits provided to members, fitness centres also use membership programmes to inform their marketing and class offerings.

Another national fitness chain, FitsGo, which has a local branch, is considering introducing their own membership system. They know, though, that the implementation of the above features will require them to put good processes and effective security in place to manage the data they collect.

The CEO of FitsGo has approached the IT management company that you work for to get some advice about how they may do this.

Your manager has asked you to provide the CEO with explanations of the following tasks to aid them in their decision-making.

© The Open Polytechnic of New Zealand

Task 1: Information management

Part 1: Four phases of the information lifecycle

Explain the data lifecycle at FitsGo: collection, organisation, processing, storage, usage, archiving and disposal. In your explanation, include reference to the confidentiality–integrity–availability (CIA) triad and the authentication–authorisation–accounting (AAA) framework.

Your explanation should be supported by at least two sources, with one source specifically from New Zealand, in addition to the course material.

(Word count guideline: 500 words)

(25 marks)

Part 2: Data governance

Regarding the case study, explain three essential aspects of data governance and three consequences of poor data governance.

(Word count guideline: 250 words)

(15 marks)

Task 2: Ethical principles and professionalism

Part 1: Real-world failure

The CEO of FitsGo would like you to present learnings from a real-world failure so that FitsGo can avoid similar issues and ensure ethical data handling.

Select **two** of the following real-world failures:

- Target data breach (2013)
- eBay data breach (2014)
- Uber data breach cover-up (2016)
- Equifax data breach (2017)
- Facebook–Cambridge Analytica scandal (2018).

For **each** real-world failure, explain the following:

- a summary of the real-world failure
- an ethical principle violated
- one issue, one consequence and how the issue could have been mitigated.

(Word count guideline: 400 words)

(20 marks)

© The Open Polytechnic of New Zealand

Part 2: IT Professionals NZ Code of Ethics

The CEO of FitsGo wants to create a culture of ethics. You have offered to present three tenets (principles) of the IT Professionals NZ Code of Ethics to the CEO of FitsGo.

For each tenet:

- explain why each tenet is relevant to FitsGo's employees
- describe an issue that may arise regarding the principle identified
- explain what could be done to mitigate the impact of the issue

(Word count guideline: 250 words)

(15 marks)

Task 3: Security concepts

Using the case study, explain the threats and vulnerabilities for three IT systems and assets. Your explanation should be supported by at least two sources, with one source specifically from New Zealand, in addition to the course material.

(Word count guideline: 300 words)

(15 marks)

Overall presentation and referencing

Marks will be awarded for overall presentation, including grammar, spelling, structure and flow, appropriate acknowledgements, citations and referencing (in APA style).

(10 marks)

© The Open Polytechnic of New Zealand

Marking schedule

Task 1: Information management Part 1: Four phases of the information lifecycle	**20–25 marks**	**16.5–19.5 marks**	**12.5–16 marks**	**10–12 marks**	**1–9.5 marks**
Explain the data lifecycle at FitsGo: collection, organisation, processing, storage, usage, archiving and disposal. In your explanation, include reference to the confidentiality–integrity–availability (CIA) triad and the authentication–authorisation–accounting (AAA) framework.	Thoroughly explains what data could be captured, organised, processed, maintained, utilised, archived and destroyed. Provides detailed justifications for why each step is necessary and how it should be implemented. Incorporates comprehensive references to the CIA triad and the AAA framework.	Clearly explains what data could be captured, organised, processed, maintained, utilised, archived and destroyed. Provides solid justifications for why each step is necessary and how it should be implemented. References the CIA triad and the AAA framework.	Adequately explains what data could be captured, organised, processed, maintained, utilised, archived and destroyed but lacks detail in one or more steps. Provides adequate justifications for why each step is necessary and how it should be implemented. References the CIA triad and the AAA framework but may lack depth in some areas.	Briefly explains what data could be captured, organised, processed, maintained, utilised, archived, and destroyed but provides insufficient detail or justifications for why each step is necessary and how it should be implemented. Provides limited references to the CIA triad and the AAA framework.	Provides minimal explanation of what data could be captured, organised, processed, maintained, utilised, archived and destroyed. Fails to justify why each step is necessary or how it should be implemented. Lacks references to the CIA triad and the AAA framework.
Task 1: Information management Part 2: Data governance	**12–15 marks**	**10–11.5 marks**	**7.5–9.5 marks**	**6–7 marks**	**1–5.5 marks**
Regarding the case study, explain three essential aspects of data governance and three consequences of poor data governance.	Thoroughly explains three essential aspects of data governance in the context of the case study. Thoroughly identifies and describes three consequences of poor data governance. Provides detailed justifications and examples for each aspect and consequence.	Clearly explains three essential aspects of data governance in the context of the case study. Clearly identifies and describes three consequences of poor data governance. Provides solid justifications and examples for each aspect and consequence.	Adequately explains three essential aspects of data governance in the context of the case study but lacks detail in one or more aspects. Identifies and describes three consequences of poor data governance but may lack depth. Provides adequate justifications and examples for each	Briefly explains two essential aspects of data governance in the context of the case study but provides insufficient detail. Identifies two consequences of poor data governance but provides limited descriptions. Offers minimal justifications and examples.	Provides minimal explanation of one to two essential aspects of data governance in the context of the case study. Fails to identify or describe three consequences of poor data governance. Lacks justifications and examples.

© The Open Polytechnic of New Zealand

			aspect and consequence.		
Task 2: Ethical principles and professionalism **Part 1: Real-world failure**	**16–20 marks**	**13–15.5 marks**	**10–12.5 marks**	**8–9.5 marks**	**1–7.5 marks**
The CEO of FitsGo would like you to present learnings from a real-world failure so that FitsGo can avoid similar issues and ensure ethical data handling. Select **two** real-world failures and explain the following: • a summary of the real-world failure • an ethical principle violated • one issue, one consequence and how the issue could have been mitigated.	Thoroughly explains two real-world failures from the list provided, including a comprehensive summary. For each failure, thoroughly identifies an ethical principle violated. Provides a detailed analysis of one issue, one consequence and how the issues could have been mitigated.	Clearly explains two real-world failures from the provided list, including a clear summary. For each failure, identifies the ethical principle violated and provides a solid analysis of one issue, one consequence and how the issue could have been mitigated.	Adequately explains two real-world failures from the provided list, including a basic summary. For each failure, identifies the ethical principle violated but may lack depth. Provides an adequate analysis of one issue, one consequence and how the issue could have been mitigated but may miss some details.	Briefly explains one or two real-world failures from the provided list but provides insufficient detail in the summary. Identifies the ethical principle violated but lacks clarity. Provides a superficial analysis of one issue, one consequence or how the issue could have been mitigated.	Provides minimal explanation of a real-world failure from the provided list. Fails to identify the ethical principle violated or provides incorrect information. Provides minimal analysis of one issue, one consequence or how the issue could have been mitigated.
Task 2: Ethical principles and professionalism **Part 2: IT Professionals NZ Code of Ethics**	**12–15 marks**	**10–11.5 marks**	**7.5–9.5 marks**	**6–7 marks**	**1–5.5 marks**
The CEO of FitsGo wants to create a culture of ethics. You have offered to present three tenets (principles) of the IT Professionals NZ Code of Ethics to the CEO of FitsGo.	Thoroughly explains three tenets of the IT Professionals NZ Code of Ethics, detailing their relevance. Thoroughly describes a potential issue related to each principle and provides comprehensive strategies for mitigating the impact of these issues.	Clearly explains three tenets of the IT Professionals NZ Code of Ethics, detailing their relevance. Clearly describes a potential issue related to each principle and provides solid strategies for mitigating the impact of these issues.	Adequately explains three tenets of the IT Professionals NZ Code of Ethics but lacks detail in some areas. Describes a potential issue related to each principle and provides adequate strategies for mitigating the impact of these issues.	Briefly explains only two tenets of the IT Professionals NZ Code of Ethics but provides insufficient detail. Describes some potential issues related to each principle but lacks clarity or depth.	Provides minimal explanation of one to two tenets of the IT Professionals NZ Code of Ethics. Fails to describe potential issues related to each principle or provides incorrect information. Provides minimal strategies for mitigating the impact of these issues.

© The Open Polytechnic of New Zealand

Task 3: Security concepts	**12–15 marks**	**10–11.5 marks**	**7.5–9.5 marks**	**6–7 marks**	**1–5.5 marks**
Using the case study, explain the threats and vulnerabilities for three IT systems and assets.	Accurately identifies three highly relevant IT systems and assets, with clear and appropriate selection from the case study. Provides a detailed and thorough explanation of the key threats and vulnerabilities for each identified system or asset, with clear links to potential threats. Explanation is well supported by two relevant sources from the course material and a New Zealand source.	Clearly identifies three relevant IT systems and assets appropriate to the case study. Provides a clear explanation of the key threats and vulnerabilities for each identified system or asset, with clear links to potential threats. Explanation is well supported by two sources from the course material and a New Zealand source.	Identifies three relevant IT systems and assets, appropriate to the case study. Provides an explanation of the key threats and vulnerabilities for each identified system or asset but lacks detail in some areas, missing some critical points. Explanation is supported by two sources from the course material and a New Zealand source.	Identifies three or fewer IT systems and assets but provides insufficient detail. Identifies some threats and vulnerabilities but lacks clarity or depth. Explanation is vaguely supported by two or one source from the course material and/or a New Zealand source.	Identifies two or one IT systems and assets. Provides minimal explanation of the IT systems and assets. Fails to identify threats and vulnerabilities or provides incorrect information. Explanation lacks or is not supported by two sources from the course material or a New Zealand source.
Presentation	**8–10 marks**	**6.5–7.5 marks**	**5–6 marks**	**4–4.5 marks**	**1–3.5 marks**
Marks will be awarded for overall presentation, including grammar, spelling, structure and flow, appropriate acknowledgements, citations and referencing (in APA style).	Presentation is flawless in all aspects: excellent grammar, logical structure, complete acknowledgements and APA formatting for citations and references, including the reference list without error.	Presentation is with minor issues in grammar or structure, acknowledgements and/or APA formatting for citations and references. Includes a reference list with only minor errors.	Presentation is satisfactory but has several noticeable issues in grammar, structure or APA formatting that moderately affect its quality. Includes a reference list with only minor errors.	Presentation has significant errors in grammar, structure and/or APA formatting. Includes a reference list with some errors.	Presentation is unsatisfactory, with major errors in all aspects, making it unreadable and lacking in academic rigour. Does not include a reference list.

© The Open Polytechnic of New Zealand